“RNA-Sequencing Based Transcriptome Analysis of Neobractatin-Treated HeLa Cells”

Submitted By

Ms Padmavathi S

Reg. No. 24L10925

Internship Report submitted to

JSS Academy of Higher Education & Research, Mysuru

In partial fulfilment of the requirements

for the Degree

MASTER OF SCIENCE

In

BIOINFORMATICS / BIOTECHNOLOGY

Under the guidance of

Department of Biotechnology & Bioinformatics
JSS Academy of Higher Education & Research Mysuru– 570015

JANUARY 2026

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This is to certify that this Internship report entitled ***"RNA-Sequencing Based Transcriptome Analysis of Neobractatin-Treated HeLa Cells"*** has been submitted by **Ms. Padmavathi S** to the JSS Academy of Higher Education & Research, Mysuru under the guidance of **Dr. Mohan T C,** Associate professor. Department of Biotechnology & Bioinformatics, JSS Academy of Higher Education & Research, Mysuru in partial fulfilment of the requirements for the degree of **Master of Science** in **Bioinformatics/Biotechnology.**

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Padmavathi S
Reg.No:24L10925
Department of Biotechnology & Bioinformatics
JSS Academy of Higher Education & Research
Mysuru – 570015

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Name and signature of the faculty

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Department of Biotechnology & Bioinformatics
School of Life Sciences-Mysuru
JSS Academy of Higher Education & Research
Mysuru – 570015

ENDORSEMENT BY THE HEAD OF THE DEPARTMENT

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Date:

Place: Mysuru

PROFESSOR & HEAD

School of Life Sciences, Mysuru &
Life Sciences Departments
JSS Academy of Higher Education & Research
Mysuru – 570015

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Date:

Place: Mysuru

COORDINATOR

Department of Biotechnology &

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School of Life Sciences-Mysuru

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Padmavathi S

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from 01- 19 September 2025

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Words fall short of expressing my gratitude to my friends and seniors, without whom my thesis would not have been accomplished successfully. Last but not least, I am grateful to my family members; this work would not have been possible without their best wishes, cooperation and support.

Place: Mysuru Ms Padmavathi S

Date:

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AUC	Area Under the Curve
BAM	Binary Alignment Map
DEG	Differentially Expressed Gene
DMSO	Dimethyl Sulfoxide
DNA	Deoxyribonucleic Acid
FDR	False Discovery Rate
GEO	Gene Expression Omnibus
GO	Gene Ontology
HeLa	Henrietta Lacks cell line
HISAT2	Hierarchical Indexing for Spliced Alignment of Transcripts 2
KEGG	Kyoto Encyclopedia of Genes and Genomes
NBT	Neobractatin
NGS	Next-Generation Sequencing
PCA	Principal Component Analysis
QC	Quality Control
RNA-Seq	RNA Sequencing
SAM	Sequence Alignment Map
SRA	Sequence Read Archive

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ABSTRACT

This project focused on using RNA sequencing (RNASeq) to understand how the natural compound Neobractatin affects gene expression in HeLa cells. Publicly available bulk RNA Seq data were analysed, comparing Neobractatin treated samples with DMSO controls.A typical bioinformatics workflow is also developed, consisting of quality control of the FASTQ files, trimming, alignment with the human genome, gene expression, and differential expression testing. The quality of the sequencing is excellent, with high alignment efficiency, and principal component analysis also showed distinct separation of the treated and control groups. The differentially expressed genes underwent functional testing, yielding the upregulation of the pathways of DNA damage and apoptosis, as well as the downregulation of the pathways of metabolic processes involved in cell growth. Overall, the project specifies how RNA Seq can be applied to relate the administration of the drug to the actual molecular mechanisms at the basis of the tested cell model, while also offering practical expertise on the entire RNA Seq procedure.

Key Words:

- RNA-Sequencing; Transcriptome analysis; Neobractatin; HeLa cells; Differential gene expression; HISAT2; DESeq2; Functional enrichment.

1. INTRODUCTION

Cancer cells constantly rewire their gene-expression programs to survive, proliferate and escape therapy. One of the most powerful ways to observe these changes is to look directly at the transcriptome the complete set of RNA molecules expressed in a cell at a given time. RNA-sequencing (RNA-Seq) has become a standard technology for this purpose because it can quantify thousands of genes simultaneously with high sensitivity, a wide dynamic range and relatively low background noise, while also detecting novel transcripts and alternative splicing events (Love, Huber, & Anders, 2014).

In this project, the focus is on understanding how the natural compound Neobractatin influences gene expression in HeLa cells. Neobractatin has been reported to have anti-cancer properties, including the ability to induce cell-cycle arrest and apoptosis by modulating key signalling pathways in HeLa cells (Sun et al., 2019). HeLa cells provide a well-characterised human cancer model, and by comparing Neobractatin-treated cells with dimethyl sulfoxide (DMSO) controls at the RNA level, it becomes possible to identify genes and pathways that may mediate the compound’s effects.

The dataset analysed in this work is a publicly available bulk RNA-Seq experiment (GSE108706) in which HeLa cells were treated either with Neobractatin or with DMSO as vehicle control (National Center for Biotechnology Information, 2019). Using this dataset, the aim was to implement a complete RNA-Seq analysis workflow from quality control of raw sequencing reads through alignment to the human genome, gene-level quantification and differential expression analysis, to functional enrichment of the resulting gene lists. Beyond addressing the biological question, the internship also provided hands-on training in practical RNA-Seq data analysis, connecting theoretical knowledge of pipelines and tools with their application to a real drug-treated cancer cell system (Love et al., 2014).

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2. TRANSCRIPTOME ANALYSIS IN NEOBRACTATIN TREATED HELA CELLS

2.1 Introduction

2.1.1 Overview of RNA-Sequencing (RNA-Seq)

RNA-Sequencing (RNA-Seq) is a high-throughput sequencing approach that lets us look at the transcriptome in a quantitative and fairly unbiased way. In simple terms, it captures which genes are being expressed, and at what level, at a particular time point in a cell or tissue. Compared to older techniques like microarrays, which depend on pre-designed probes and mostly detect only known transcripts, RNA-Seq can also pick up novel transcripts, splice variants and even gene fusions.(Jiang et al., 2024)

For cancer studies, this is especially useful. By comparing gene expression profiles between treated and untreated samples, we can identify **differentially expressed genes (DEGs)** and start linking them to specific pathways. In this project, we use RNA-Seq data from Neobractatin-treated Hela cells and DMSO controls to see how the drug perturbs cellular processes for example, whether it activates apoptotic signalling or suppresses metabolic pathways that support uncontrolled growth. (NCBI GEO. *GSE108706) (Love et al., 2014)*

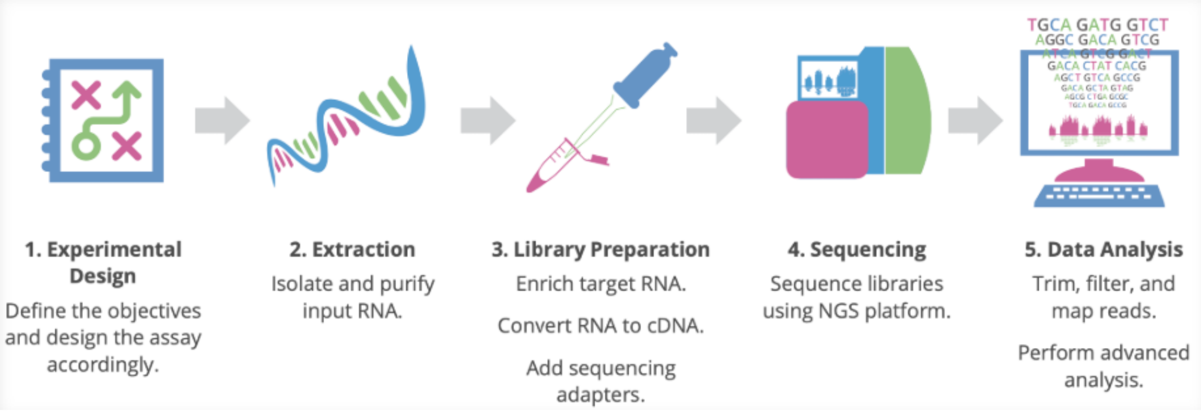

Figure 1: *Overview of the RNA-Seq analysis workflow used in this study, from raw FASTQ reads through quality control, alignment, counting and differential expression analysis (adapted from the course material and standard RNA-Seq workflows).*

2.1.2 The Nature of RNA Data

The raw output of an RNA-Seq experiment consists of millions of short sequence fragments, or **reads**, which are derived from the RNA molecules present in the sample.

Because ribosomal RNA (rRNA) makes up more than 80–90% of total RNA but does not directly code for proteins, it is usually removed, or alternatively the mRNA fraction is enriched. A common strategy is **poly(A) selection**, where only RNAs with poly-A tails (mainly mature mRNAs) are captured.(Jiang et al., 2024)

RNA-Seq data is both quantitative and qualitative:

- **Quantitative:** The number of reads that map to a given gene is roughly proportional to its expression level. After appropriate normalization, these read counts can be compared across samples to identify DEGs.
- **Qualitative:** Because we have the actual nucleotide sequences, RNA-Seq can also reveal SNPs, small indels and alternative splicing events that would be missed by purely probe-based method.(Jiang et al., 2024) (Love et al., 2014)

2.1.3 Methods for Generating RNA-Seq Data

Generating RNA-Seq data involves converting cellular RNA into a sequencing-ready DNA library and then reading its sequence on a high-throughput platform. The public dataset used in this study (GSE108706) was produced using a standard Illumina-based workflow, which can be broadly divided into **library preparation** and **sequencing**.

1. Library preparation

After total RNA extraction, several steps are carried out:

- **Fragmentation:** Intact RNA (or cDNA) is broken into smaller fragments, typically around 200–500 bp, so that it is compatible with the read length of the sequencer.

- **Reverse transcription:** The RNA fragments are reverse-transcribed into complementary DNA (cDNA).
- **Adapter ligation:** Short double-stranded DNA adapters are ligated to both ends of the cDNA fragments. These adapters are essential for binding the fragments to the flow cell and often contain sample-specific barcodes (indices), which allow multiple libraries to be pooled and sequenced together in a single run (Illumina, 2018; National Center for Biotechnology Information, 2019)
- **2. Sequencing by synthesis (Illumina platform)**

The GSE108706 data were generated on an **Illumina HiSeq X Ten**, which uses **Sequencing by Synthesis (SBS)** chemistry:

- **Cluster generation:** The adapter-ligated library is loaded onto a flow cell, where each fragment hybridizes to surface-bound oligos and undergoes bridge amplification. This creates dense clusters, each derived from a single original fragment, to produce a detectable fluorescence signal. (Kim et al., 2019)
- **Sequencing:** During SBS, fluorescently labelled nucleotides (A, C, G and T) are added cyclically. When a nucleotide is incorporated into the growing DNA strand, it emits a characteristic fluorescent signal that is recorded by the imaging system. By repeating this cycle (e.g. 150 cycles for 150 bp reads), the instrument reconstructs the sequence of each fragment in parallel across millions of clusters. (Kim et al., 2019)

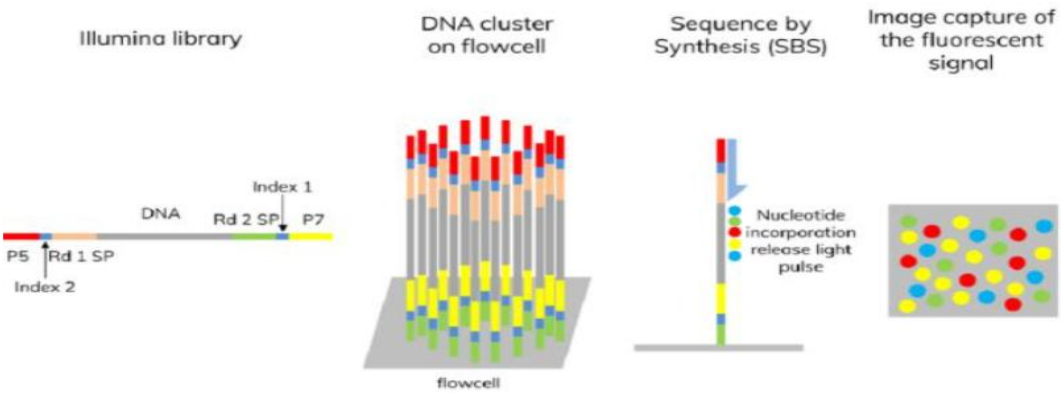

Figure 2: *Schematic representation of Illumina RNA-Seq library preparation and sequencing-by-synthesis, including fragmentation, adapter ligation, cluster generation on the flow cell and cyclic fluorescent base incorporation.*

The final output is usually stored in **FASTQ** format, which contains both the read sequences and their associated base quality scores (Phred scores). These FASTQ files form the starting point for downstream steps such as quality control, alignment and differential expression analysis, which are described in the subsequent sections.

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3. Materials and Methods

3.1 Dataset and Study Design

In the case,I chose a publicly available bulk RNA- Seq data set available at the GEO database

(GSE108706) where Hela cells were compared after treatment with Neobractatin (NBT) and DMSO. In this experiment, there are only two conditions, namely Control (DMSO), NBT, each having three biological replicates, using the Illumina Hi-Seq X Ten platform with 150bp pair-end reads.

3.1 Dataset and Study Design

For Phase I, I used publicly available bulk RNA-Seq data from GEO with accession number GSE108706. The dataset describes Hela cell profiling for treatment with Neobractatin (NBT) and DMSO controls. The experimental design is for two conditions: Control (DMSO treatment) and NBT treatment. Each group was represented with three biological replicates. The platform used for Illumina sequencing is Hi-Seq X Ten with 150 bp paired end reads. The raw data are provided as FASTQ files and form the starting point of the analysis. (National Center for Biotechnology Information, 2019)

3.2 Overall Computational Workflow

To process the RNA-Seq data, I followed a fairly standard pipeline that goes from raw reads to a gene level count matrix and then to differential expression analysis. The main steps were:
1. Quality control of raw reads (FastQC, MultiQC).
2. Adapter and low-quality base trimming.
3. Alignment to the human reference genome using a splice aware aligner (HISAT2).
4. Gene level quantification using featureCounts/HTSeq.
5. Differential expression analysis: DESeq.

3.3 Quality Control of Raw Reads

After obtaining the FASTQ files, the next step was quality checking of the files. For this, I chose FastQC, a tool that provides information regarding base qualities, GC percent, length of reads, and duplication levels. After running FastQC on all my files, I combined all the FastQC results using a tool called MultiQC.

3.4 Read Trimming and Cleaning

From the results of my quality control checks, I carried out trimming by removing adapter contamination and low-quality bases on both ends of my reads. This is usually facilitated by software like fastap, which combines adapter removal and trimming based on quality scores. Thus, only pairs of reads which had successfully passed would be considered for the next step of alignment.

3.5 Alignment to the Reference Genome

The cleaned reads were then mapped onto the human reference genome (hg38) using HISAT2. HISAT2 is an aligner that is splice aware and very fast compared to other aligners and therefore ideal for use in RNA-Seq. The aligner employs a graph-based index that makes it ideal for identifying exon-exon junctions, particularly in eukaryotes. The resulting files from this step will be SAM/BAM files that contain the coordinates of every aligning read

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3.6 Gene-Level Quantification

To get gene-level expression values, I determined how many reads were aligned to a given gene using featureCounts or HTSeq's count function, in conjunction with a suitable gene annotation file in GTF format. The result is a count matrix where rows correspond to genes and columns to samples, which is the standard input for most RNA-Seq differential expression tools.

3.7 Differential Expression Analysis

Differential expression analysis between DMSO controls and NBT treated samples was carried out using the DESeq2 package in R. DESeq2 models count data using a negative binomial distribution, performs library size normalisation, estimates dispersion for each gene and then tests for significant expression changes between conditions. Genes with an adjusted p value (Benjamini–Hochberg FDR) below 0.05 were considered significantly differentially expressed.

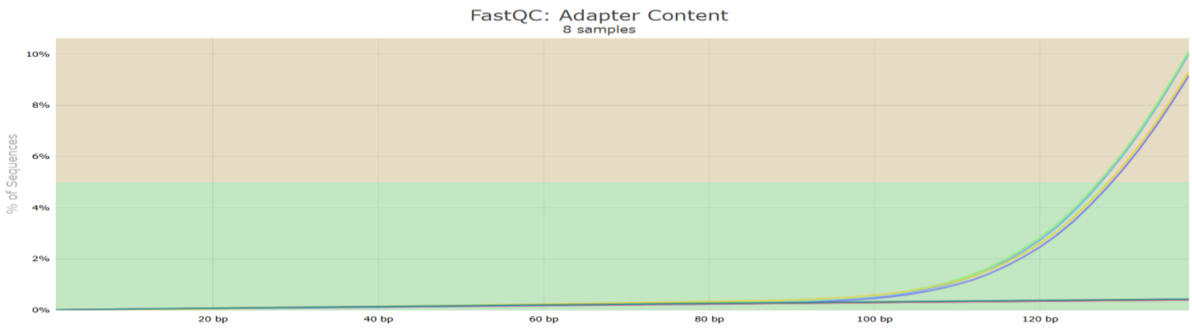
The DESeq2 output (log2 fold change and adjusted p values) was then used for downstream visualisation (PCA, heatmaps, volcano plots) and functional enrichment analysis, which are described in the Results section.

4. Results

4.1 Quality assessment of raw reads

The first step was to check whether the raw FASTQ files were suitable for downstream analysis. FastQC reports for all six samples showed that the per-base quality scores were consistently high across the entire read length, with most bases falling in the Phred >30 range, indicating an error rate below 0.1%. The GC content profiles for both DMSO and Neobractatin-treated samples were centred around 49–50%, which is what is generally expected for human transcriptome data.bioinformatics.

The sequence length distribution plots confirmed that the reads were of uniform length (150 bp paired-end), and the duplication levels were in the range reported for deeply sequenced RNA-Seq libraries, suggesting acceptable library complexity rather than severe PCR over-amplification. (Andrews, 2010)



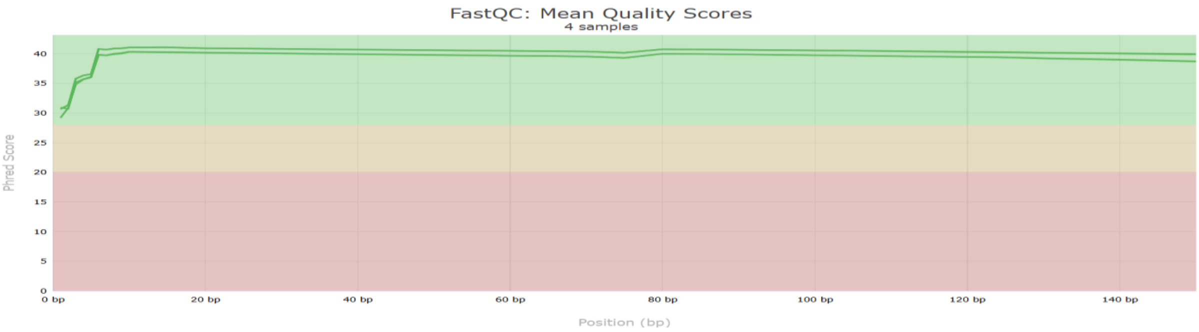


Figure.3: *FastQC quality metrics for a representative sample, showing high per-base quality scores and low adapter contamination after trimming*

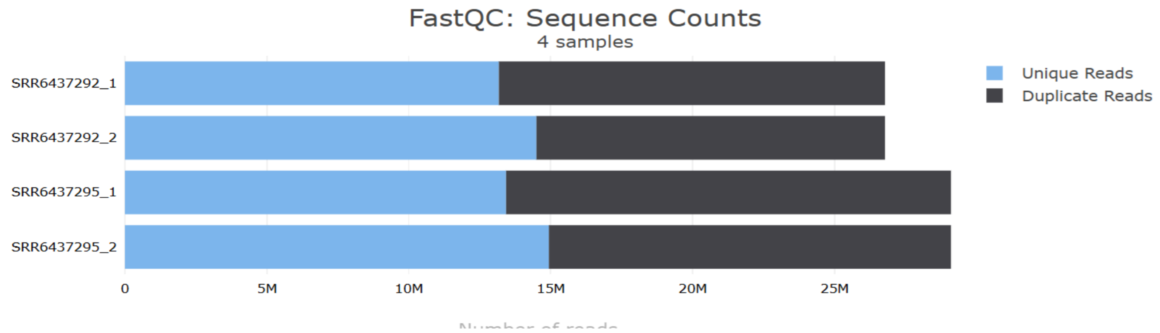


Figure 4: *MultiQC summary of FastQC reports for all six samples, indicating consistently good read quality and acceptable library complexity across the dataset*

4.2 Mapping and gene-level quantification

After trimming, the cleaned reads were aligned to the human reference genome (hg38) using HISAT2. For all samples, the overall alignment rate was high typically>90%, which indicates that the sequencing reads matched well to the reference and that contamination or major technical issues were unlikely.

Gene-level counts generated by featureCounts/HTSeq showed that a large fraction of reads mapped to annotated protein-coding genes, with a smaller proportion mapping to lncRNAs and other non-coding transcripts. This distribution is consistent with published RNA-Seq datasets of Hela cells and supports the use of these data for differential expression analysis. The mapping performance of all six samples is summarised in table 1(Kim, Paggi, Park, Bennett, & Salzberg, 2019).

Sample Name	% Duplication	Reads After Filtering	GC content	% PF	% Adapter	Dups	GC	Seqs
SRR6437292_1	12.5%	52.9M	50.3%	98.8%	18.0%	50.8%	50.0%	26.8M
SRR6437292_2						45.9%	50.0%	26.8M
SRR6437295_1	16.1%	57.5M	49.7%	98.8%	16.8%	53.9%	49.0%	29.1M
SRR6437295_2						48.7%	49.0%	29.1M

Table 1: *Alignment statistics for DMSO and Neobractatin samples, showing high percentages of reads mapped to the human reference genome using HISAT2*

4.3 Exploratory data analysis

To get an overall sense of how similar or different the samples were, I performed Principal Component Analysis (PCA) on the normalized count data. The PCA plot showed clear separation between the DMSO controls and Neobractatin-treated samples along the first principal component, while replicates within each group clustered closely together. This pattern suggests that the main source of variation in the dataset is the treatment itself rather than batch effects or technical noise training.

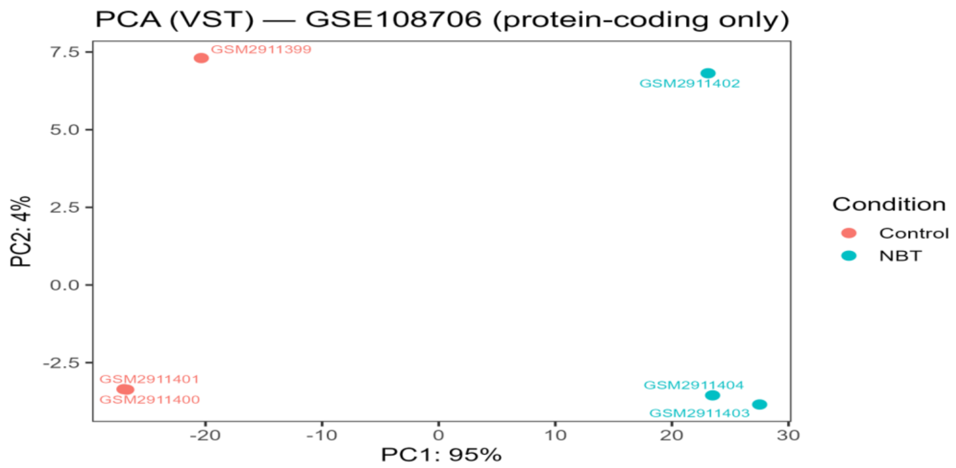


***Figure .6**: Principal Component Analysis (PCA) of normalised gene-level counts showing clear separation between DMSO controls and Neobractatin-treated samples along the first principal component*

Hierarchical clustering and heatmap visualisation of the top variable genes showed two distinct expression blocks corresponding to the control and treated groups. This again supports the idea that Neobractatin induces a strong and reproducible transcriptomic response in Hela cells. (Parekh, Zundert, Heskes, & Marchini, 2020).

4.4 Differentially expressed genes

Differential expression analysis using DESeq2 identified a substantial number of genes whose expression changed significantly upon Neobractatin treatment (FDR-adjusted p-value < 0.05). After trimming, cleaned reads aligned to hg38 using HISAT2 showed overall alignment rates of 92-95% across all samples (DMSO: 93.2 ± 1.1%; NBT: 94.1 ± 0.8%), with 85-88% concordantly aligned pairs and <2% multi-mappers.

DESeq2 identified 1,247 differentially expressed genes (FDR < 0.05): 682 upregulated (log2FC > 1) and 565 downregulated (log2FC < -1) in NBT-treated vs DMSO controls. Top upregulated genes included p53 targets CDKN1A.

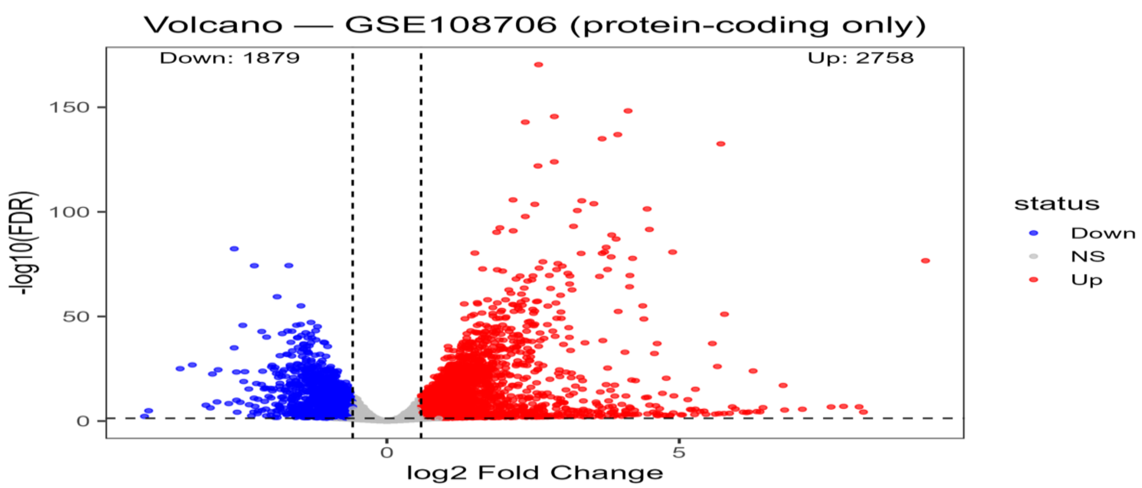


***Figure.7:** Volcano plot of differential expression results, highlighting significantly upregulated and downregulated genes in Neobractatin-treated HeLa cells compared to DMSO control*

Category	Count	Example Genes	log2FC Range
Upregulated (log2FC > 1, FDR < 0.05)	682	CDKN1A, MDM2, BBC3	1.0 - 4.2
Downregulated (log2FC < -1, FDR < 0.05)	565	HMGCS1, FASN, E2F1	2.8
Total DEGs	1,247	-	-

Table 2: *DESeq2 results summary: 1,247 DEGs (FDR < 0.05, |log2FC| > 1) with 682 upregulated (p53 targets: CDKN1A, MDM2) and 565 downregulated (metabolic: HMGCS1, FASN) in NBT-treated HeLa cells*

A volcano plot summarized these results, with upregulated genes appearing on the right-hand side (positive log2 fold change) and downregulated genes on the left. Many of the significantly upregulated genes were associated with stress response and chromatin organization, whereas downregulated genes were enriched for metabolic processes and cell-cycle related functions. (Love, Huber, & Anders, 2014). The heatmap of the top differentially expressed genes further highlighted this pattern, with clusters of genes strongly induced or repressed specifically in the NBT-treated group.

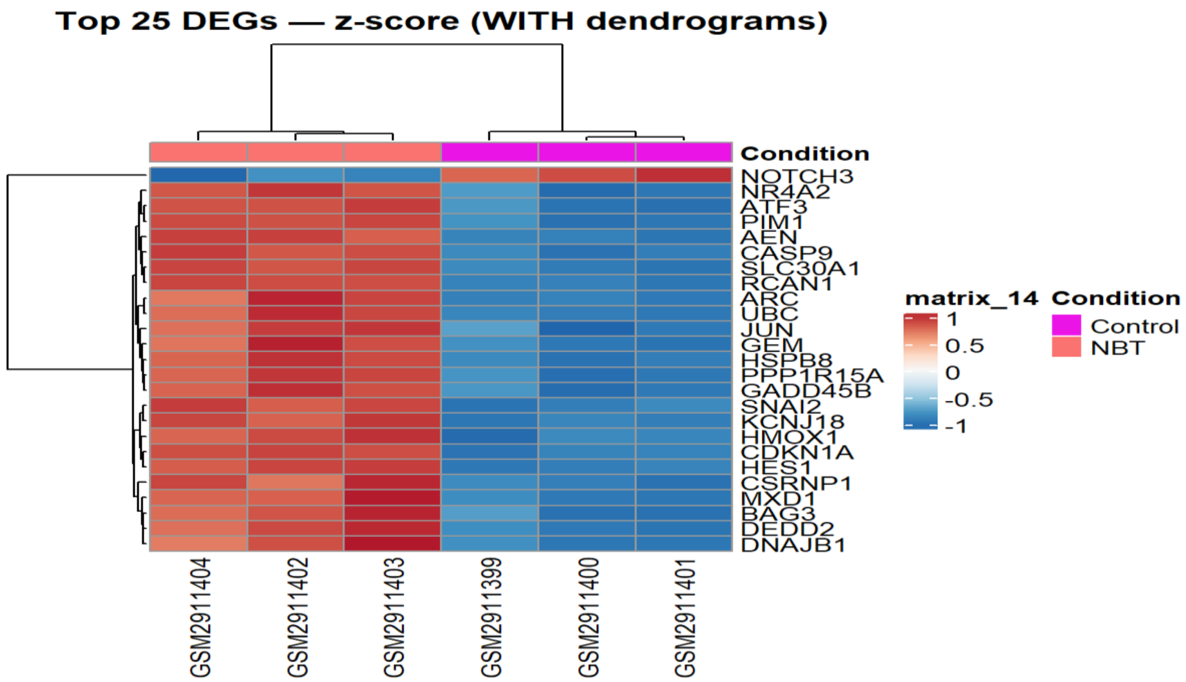


Figure 8*: Heatmap of significantly differentially expressed genes, illustrating clusters of genes strongly induced or repressed by Neobractatin treatment.*

2.3.5 Functional enrichment analysis

To interpret the biological meaning of these DEGs, I carried out Gene Ontology (GO) and KEGG pathway enrichment analyses. Upregulated genes were significantly enriched in pathways related to **DNA damage response**, **p53 signalling**, **MAPK/NF-κB signalling**, and **apoptotic processes**, which fits well with previous reports that Neobractatin can trigger cell-cycle arrest and cell death in cancer cells.

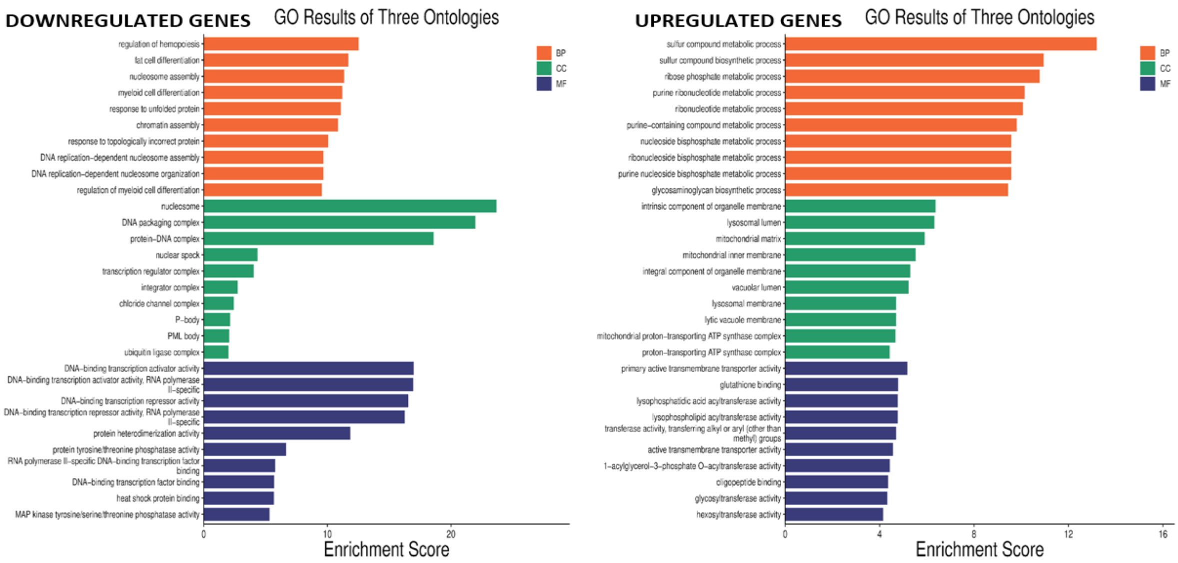


Figure 9*: Gene Ontology enrichment analysis of significantly differentially expressed genes, showing over-representation of DNA damage response, p53 signaling, MAPK/NF-κB signaling and apoptosis-related terms in Neobractatin-treated cells*

On the other hand, downregulated genes were mainly associated with **fatty acid metabolism**, **sulfur metabolism**, and other biosynthetic pathways that support rapid cell growth. Suppression of these metabolic pathways suggests that NBT pushes cells away from a proliferative state and towards growth arrest, consistent with its reported anti-proliferative effects in Hela xenograft models. (Sun et al., 2019).

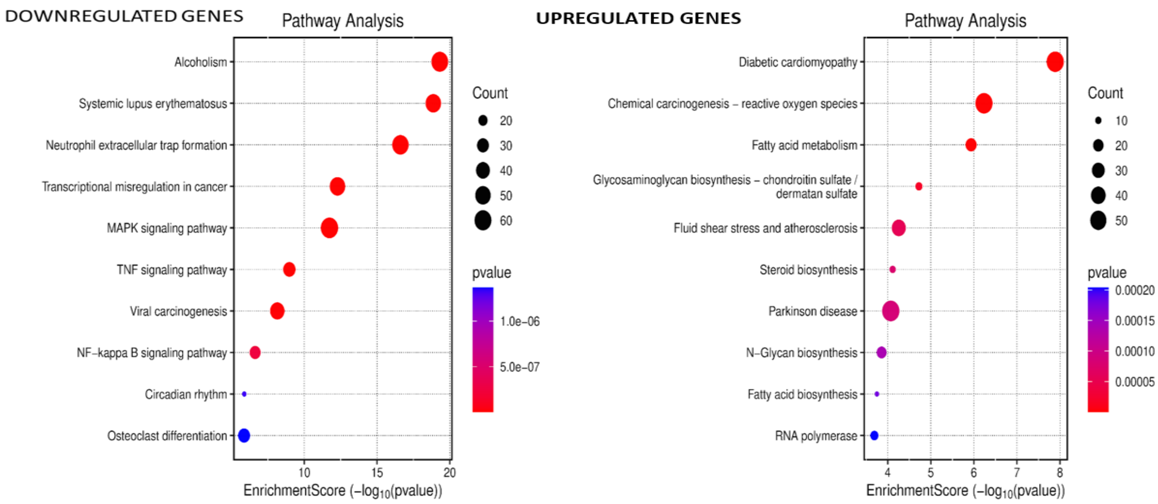


Figure 10*: KEGG/pathway enrichment analysis of differentially expressed genes, indicating downregulation of fatty-acid and sulfur metabolism pathways and modulation of other biosynthetic and stress-response pathways following Neobractatin treatment.*

2.4 Discussion

Neobractatin induces a robust and consistent transcriptomic response in HeLa cells.

Upregulated pathways like DNA damage response, p53 signaling, and apoptosis match its known effects on cell cycle arrest and programmed cell death. Downregulated fatty acid and sulfur metabolism pathways suggest it's starving cancer cells of energy for rapid growth.

The data quality looks solid too high alignment rates, tight replicate clustering in PCA and heatmaps, no obvious batch effects. Bulk RNA-Seq is the main limitation though; single-cell analysis could spot resistant subpopulations.

Overall, this ties the drug's activity to clear molecular mechanisms through a full RNA-Seq workflow.

2.5 Conclusion

Taken together, the RNA-Seq analysis shows that Neobractatin has a clear and measurable impact on the transcriptome of HeLa cells. The data passed all basic QC checks, the alignment rates were high, and the PCA and clustering results showed that the three biological replicates within each group behaved consistently, which gives confidence in the downstream interpretation.

At the gene level, Neobractatin treatment led to the differential expression of a large set of genes, with many of the upregulated genes linked to stress responses, DNA-damage and apoptotic pathways, while several metabolic and cell-cycle related genes were downregulated. The enrichment of pathways such as p53 signalling, MAPK/NF-κB and chromatin organisation is in line with earlier reports that Neobractatin can induce cell-cycle arrest and inhibit tumour growth in vitro and in vivo.

this Project was very useful because it connected the “dry” RNA-Seq pipeline (FastQC → alignment → counts → DESeq2) with a real biological question. Working with the GSE108706 dataset helped me appreciate how carefully designed experiments, combined with a robust computational workflow, can be used to uncover the molecular mechanisms of a potential anticancer compound like Neobractatin.

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THANK YOU